

# UNIVERSITY EXAMINATION

## PHYSICS: THERMODYNAMICS & OPTICS

Time Allowed: 2 Hours

Maximum Marks: 50

### GENERAL INSTRUCTIONS:

- 1. Answer all questions according to section guidelines.
- 2. Figures to the right indicate full marks allocated to each question.

### SECTION A - MULTIPLE CHOICE QUESTIONS

*Select the correct alternative for each question.*

- Q1.** Which of the following thermodynamics laws defines temperature? [1]  
A. First Law of Thermodynamics                      B. Zeroth Law of Thermodynamics  
C. Second Law of Thermodynamics                      D. Third Law of Thermodynamics
- Q2.** What is the efficiency of a Carnot engine operating between temperatures 300 K and 600 K? [1]  
A. 25%                      B. 50%  
C. 75%                      D. 100%
- Q3.** The root mean square speed of gas molecules is directly proportional to: [1]  
A. Absolute temperature  $T$                       B. Square root of absolute temperature  $\sqrt{T}$   
C. Pressure  $P$                       D. Inverse of absolute temperature  $1/T$
- Q4.** In a Young's double-slit experiment, if the distance between slits is halved and the distance to the screen is doubled, the fringe width will: [1]  
A. Remain unchanged                      B. Be halved  
C. Be doubled                      D. Be quadrupled
- Q5.** Polarization of light proves that light waves are: [1]  
A. Longitudinal waves                      B. Transverse waves  
C. Electromagnetic scalar waves                      D. Mechanical stationary waves

### SECTION B - SHORT ANSWER QUESTIONS

*Answer any five questions. (Short responses required)*

- Q6.** State and explain the First Law of Thermodynamics, highlighting its relationship with the conservation of energy. [3]
- Q7.** Explain the physical significance of Entropy and state the Clausius statement of the Second Law of Thermodynamics. [3]

- 
- Q8.** Write down van der Waals equation for real gases and explain the correction terms for pressure and volume. [3]
- Q9.** Differentiate between Fresnel and Fraunhofer classes of diffraction. [3]
- Q10.** State Brewster's Law of polarization and derive the condition relating the polarizing angle to the refractive index. [3]

### SECTION C - NUMERICAL / PROBLEM QUESTIONS

*Solve all numerical problems with proper step-by-step calculations.*

- Q11.** A Carnot engine absorbs 2000 J of heat from a reservoir at 500 K and rejects heat to a sink at 300 K. Calculate (a) the efficiency of the engine, (b) the amount of heat rejected to the sink, and (c) the total work done by the engine. [5]
- Q12.** Calculate the root mean square velocity of oxygen molecules ( $M = 32 \text{ g/mol}$ ) at a room temperature of 27 degrees Celsius. Given  $R = 8.314 \text{ J/(mol}\cdot\text{K)}$ . [5]
- Q13.** In a Newton's rings experiment, the diameter of the 10th dark ring changes from 1.40 cm to 1.27 cm when a liquid is introduced between the lens and the glass plate. Calculate the refractive index of the liquid. [5]
- Q14.** Light of wavelength 589 nm is incident normally on a slit of width 0.025 mm. Calculate the angular position of the first minimum and the width of the central maximum on a screen placed 2.0 m away. [5]

### SECTION D - LONG ANSWER QUESTIONS

*Answer the following comprehensive question in detail.*

- Q15.** Explain the principle of operation of a Carnot heat engine. Derive the expression for the efficiency of a Carnot engine in terms of source and sink absolute temperatures. Also, state and prove Carnot's theorem regarding the efficiency of heat engines. [10]

--- OR ---

None

[10]

---

## CONFIDENTIAL - OFFICIAL EVALUATION SCHEME & ANSWER KEY

### SECTION A - Multiple Choice Questions - Solutions

**Q1. (MCQ, 1M) [Topic: Chapter 1: Zeroth & First Laws of Thermodynamics, Work and Heat | Level: Remember]**

**Answer/Rubric:** B. Zeroth Law of Thermodynamics - [The Zeroth law of thermodynamics deals with thermal equilibrium and forms the basis for temperature measurement.]

**Q2. (MCQ, 1M) [Topic: Chapter 2: Second Law, Heat Engines, Carnot Cycle, and Entropy | Level: Apply]**

**Answer/Rubric:** B. 50% - [Efficiency =  $1 - (T_c / T_h) = 1 - (300 / 600) = 1 - 0.5 = 0.5$  or 50%.]

**Q3. (MCQ, 1M) [Topic: Chapter 3: Kinetic Theory of Gases & Mean Free Path | Level: Remember]**

**Answer/Rubric:** B. Square root of absolute temperature  $\sqrt{T}$  - [RMS speed  $v_{rms} = \sqrt{3RT/M}$ , which shows it is directly proportional to the square root of absolute temperature.]

**Q4. (MCQ, 1M) [Topic: Chapter 4: Wave Motion, Superposition Principle & Interference of Light | Level: Apply]**

**Answer/Rubric:** D. Be quadrupled - [Fringe width  $\beta = (\lambda * D) / d$ . If D becomes 2D and d becomes d/2, new  $\beta = (\lambda * 2D) / (d/2) = 4 * ((\lambda * D)/d)$ , meaning it quadruples.]

**Q5. (MCQ, 1M) [Topic: Chapter 5: Diffraction & Polarization | Level: Remember]**

**Answer/Rubric:** B. Transverse waves - [Only transverse waves can be polarized, as polarization involves restricting the oscillation direction perpendicular to the direction of propagation.]

### SECTION B - Short Answer Questions - Solutions

**Q6. (Short Answer, 3M) [Topic: Chapter 1: Zeroth & First Laws of Thermodynamics, Work and Heat | Level: Understand]**

**Answer/Rubric:** The First Law of Thermodynamics states that energy cannot be created or destroyed, only transformed from one form to another. Mathematically,  $dU = dQ - dW$ , where dU is the change in internal energy, dQ is heat added, and dW is work done by the system. It is essentially the law of conservation of energy applied to thermodynamic systems.

**Q7. (Short Answer, 3M) [Topic: Chapter 2: Second Law, Heat Engines, Carnot Cycle, and Entropy | Level: Understand]**

**Answer/Rubric:** Entropy is a measure of the unavailability of a system's thermal energy for conversion into mechanical work, often interpreted as the degree of disorder. Clausius statement: Heat cannot spontaneously flow from a colder body to a warmer body without the intervention of an external agency.

**Q8. (Short Answer, 3M) [Topic: Chapter 3: Kinetic Theory of Gases & Mean Free Path | Level: Understand]**

**Answer/Rubric:**  $(P + a/V^2)(V - b) = RT$ . The pressure correction ( $a/V^2$ ) accounts for intermolecular attractive forces which reduce the wall collision force. The volume correction (b) accounts for the finite physical size occupied by the gas molecules themselves.

**Q9. (Short Answer, 3M) [Topic: Chapter 5: Diffraction & Polarization | Level: Analyze]**

**Answer/Rubric:** In Fresnel diffraction, either the source of light or the screen (or both) is at a finite distance from the obstacle/aperture, resulting in spherical or cylindrical wavefronts incident. In Fraunhofer diffraction, both the source and the screen are effectively at an infinite distance, meaning plane wavefronts are involved.

**Q10. (Short Answer, 3M) [Topic: Chapter 5: Diffraction & Polarization | Level: Apply]**

**Answer/Rubric:** Brewster's Law states that when unpolarized light is incident at a specific polarizing angle ( $p$ ) on a transparent surface, the reflected light is completely polarized perpendicular to the plane of incidence, and the reflected and refracted rays are mutually perpendicular. Equation:  $\tan(p) = n_2 / n_1$ .

### SECTION C - Numerical / Problem Questions - Solutions

**Q11. (Numerical, 5M) [Topic: Chapter 2: Second Law, Heat Engines, Carnot Cycle, and Entropy | Level: Apply]**

**Answer/Rubric:** (a) Efficiency  $\eta = 1 - (T_c / T_h) = 1 - (300 / 500) = 0.4$  or 40%. (b) Since  $\eta = W / Q_h$ ,  $W = 0.4 * 2000 = 800$  J. Heat rejected  $Q_c = Q_h - W = 2000 - 800 = 1200$  J.

**Q12. (Numerical, 5M) [Topic: Chapter 3: Kinetic Theory of Gases & Mean Free Path | Level: Apply]**

**Answer/Rubric:** Convert T to Kelvin:  $T = 27 + 273 = 300$  K.  $M = 0.032$  kg/mol. Formula:  $v_{rms} = \sqrt{3RT / M} = \sqrt{3 * 8.314 * 300 / 0.032} = \sqrt{7482.6 / 0.032} = \sqrt{233831.25}$  approx 483.56 m/s.

**Q13. (Numerical, 5M) [Topic: Chapter 4: Wave Motion, Superposition Principle & Interference of Light | Level: Analyze]**

**Answer/Rubric:** For air,  $D_n^2 = 4nR\lambda$ . For liquid,  $D_n^2(\text{liquid}) = (4nR\lambda) / \mu$ . Therefore,  $\mu = D_n^2(\text{air}) / D_n^2(\text{liquid}) = (1.40)^2 / (1.27)^2 = 1.96 / 1.6129$  approx 1.215.

**Q14. (Numerical, 5M) [Topic: Chapter 5: Diffraction & Polarization | Level: Apply]**

**Answer/Rubric:** Given  $\lambda = 589 \times 10^{-9}$  m,  $a = 0.025 \times 10^{-3}$  m,  $D = 2.0$  m. For first minimum:  $a \sin(\theta) = \lambda$   
 $\Rightarrow \sin(\theta) = (589 \times 10^{-9}) / (2.5 \times 10^{-5}) = 0.02356$  rad. Angular position  $\theta$  approx 1.35 degrees. Width of central maximum  $= 2 * D * \lambda / a = 2 * 2.0 * (589 \times 10^{-9}) / (2.5 \times 10^{-5}) = 0.09424$  m = 9.42 cm.

## SECTION D - Long Answer Questions - Solutions

**Q15. (Long Answer, 10M) [Topic: Chapter 2: Second Law, Heat Engines, Carnot Cycle, and Entropy | Level: Evaluate]**

**Answer/Rubric:** A Carnot engine operates on a reversible cycle consisting of two isothermal and two adiabatic processes. Derivation:  $Q_1 = RT_1 \ln(V_2/V_1)$  and  $Q_2 = RT_2 \ln(V_3/V_4)$ . Using adiabatic relations,  $V_2/V_1 = V_3/V_4$ , hence  $\eta = 1 - (Q_2/Q_1) = 1 - (T_2/T_1)$ . Carnot's Theorem states that no heat engine working between two temperatures can be more efficient than a reversible Carnot engine operating between the same two temperatures. Proof involves coupling a hypothetical engine of higher efficiency with a reversed Carnot engine to violate the Second Law of Thermodynamics (Clausius or Kelvin-Planck statement).